



**FA7155 six-dimensional force sensor**

## **user's manual**

(Version: 25.04.b)

## Preface

This manual details the FA7155 six-axis force sensor, covering its functions, dimensions, usage and maintenance, communication interfaces, and precautions. Before using this product, please read this manual carefully and follow the correct operating procedures to prevent malfunctions or safety incidents caused by improper operation.

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### ● For Your Attention :

- (1) Minor color variations may occur between batches of products. Due to packaging or transportation reasons, some products may exhibit slight scratches on the surface, which does not affect product performance.
- (2) If the user uses this sensor model in special environments such as strong magnetic fields, electromagnetic radiation, or corrosive chemical environments, please contact the after-sales service.
- (3) The company shall not be liable for any economic losses resulting from the operation or use of sensors in ways not specified in the user manual.
- (4) The Company shall not be liable for any sensor damage caused by force majeure (e.g., earthquakes, tsunamis, snow disasters) or third-party actions.
- (5) The warranty period of this product is 12 months from the date of delivery.

# Catalogue

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# 1. Product Introduction

The FA7155 six-axis force sensor is a strain-based multi-axis force sensor that can be matched with most collaborative arm end-effector products on the market. It features easy installation and quick operation. This product has been widely applied in various fields such as intelligent manufacturing, automated testing, and aerospace, demonstrating excellent reliability and stability.

The FA7155 six-axis force sensor simultaneously measures forces and torques in three directions. Featuring an integrated high-precision measurement circuit, it tracks real-time strain gauge signal variations under load. Using a decoupling matrix, the sensor converts these signals into instantaneous force and torque measurements, which are then displayed on the user interface.

During installation, attach the sensor's fixed end to the user's device and connect its tool end to the user's custom tool. For optimal measurement accuracy of the six-axis force sensor, follow the instructions in the user manual for proper installation.

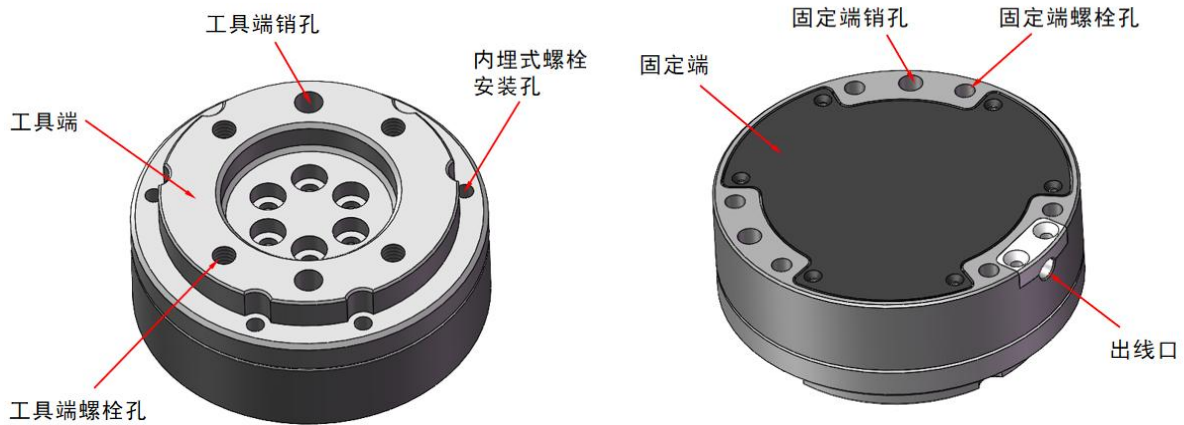


Figure 1 Schematic Diagram of the Sensor Appearance

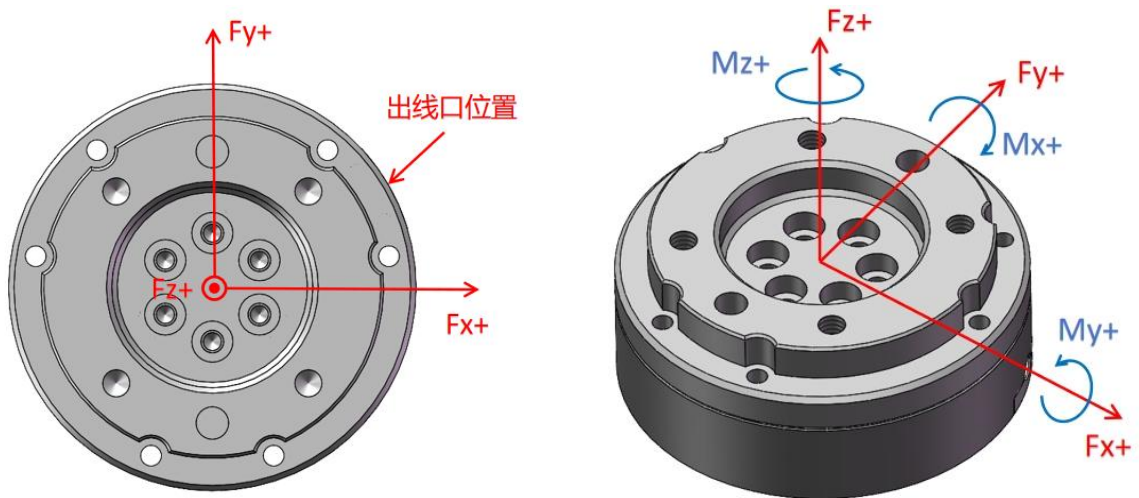


Figure 2 Schematic Diagram of Sensor Coordinate System and Force (moment) Direction

## 2. Installation Instructions

### 2.1 Sensor Installation Procedure

To ensure uniform force distribution among sensor components and at interfaces connecting the sensor to external robotic arms and custom tools, bolts should be tightened using the 'counter-rotating tightening' method, following the sequence shown in Figure 4.

If the screw hole position at the end of the user's robotic arm does not align with the sensor's mounting screw hole during installation, follow these steps to use the adapter flange:

- (1) The fixed-end adapter flange is mounted to the user equipment's end using four M6 bolts and a  $\Phi 6$  pin.
- (2) The pin hole of the six-dimensional force sensor is placed on the fixed end of the adapter flange.
- (3) Use an internal hex wrench to tighten the bolt through the embedded screw mounting hole, applying the 'counterclockwise tightening' method (as shown in Figure 4).

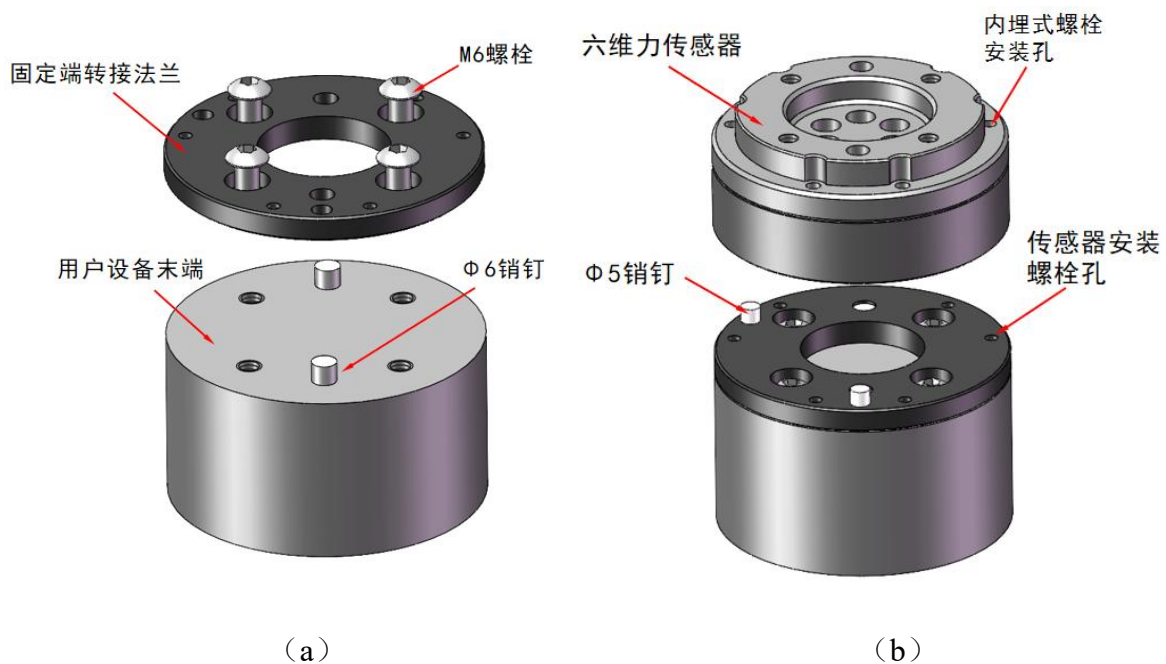


Figure 3 (a) and (b) Schematic Diagram of Sensor Installation

- (4) Connect custom tools (e.g., grinding heads, drills, etc.) to the sensor's tool end using M6 bolts.

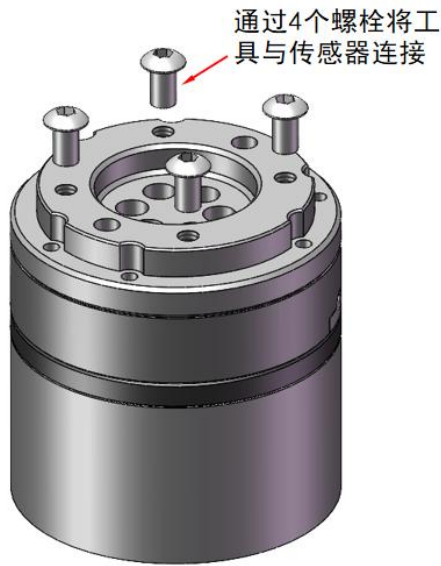


Figure 3 (c) Schematic Diagram of Tool End Fixation

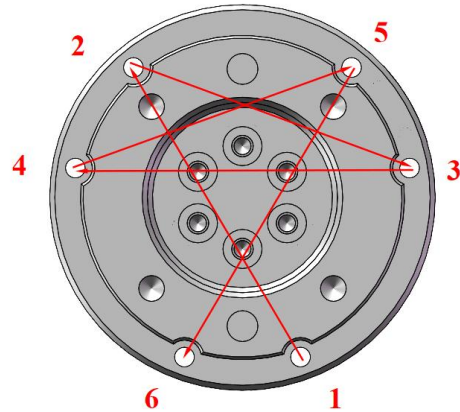


Figure 4 Sequence of Bolt Tightening When the Fixed End Is Secured

## 2.2 Cables and Electrical Interfaces

The cable used in this series of products is a high-flexibility shielded four-core cable. At the time of factory delivery, one end of the cable is connected to the inside of the sensor and cannot be disassembled. The cable is fixed at the outlet with a crimping tool. During use, it is necessary to arrange the cable path reasonably to avoid excessive bending that may damage the cable and affect the sensor's communication.

Consider a sensor with RS485 communication as an example. The sensor's output terminal has four wires: the power supply line VCC (red), the ground line GND (black), and the RS485 communication lines 485\_A (yellow) and 485\_B (green), as shown in Figure 5. When connecting the cables, always follow the color coding to prevent signal readout failures or sensor damage caused by incorrect wiring.



Figure 5 Wiring Diagram of Six-Dimensional Force Sensor

### 3. Product Technical Specifications

#### 3.1 Product Model Specifications

| Range                              |     | Product model      |
|------------------------------------|-----|--------------------|
|                                    |     | FA7155             |
| <b>F<sub>x</sub> F<sub>y</sub></b> | N   | 200                |
| <b>F<sub>z</sub></b>               | N   | 200                |
| <b>M<sub>x</sub> M<sub>y</sub></b> | Nm  | 7.5                |
| <b>M<sub>z</sub></b>               | Nm  | 7.5                |
| <b>Diameter</b>                    | mm  | 75                 |
| <b>Altitude</b>                    | mm  | 31.5 Sensor body   |
| <b>Material quality</b>            |     | Alufer             |
| <b>Surface preparation</b>         |     | Anodic oxidation   |
| <b>Anti-G</b>                      | %   | 300                |
| <b>Resolution ratio</b>            | %FS | 0.03               |
| <b>Accuracy</b>                    | %FS | 0.5                |
| <b>Repeatability precision</b>     | %FS | 0.1                |
| <b>Weight</b>                      | kg  | ≤0.3               |
| <b>Frequency of sampling</b>       | Hz  | 250 (customizable) |
| <b>Service voltage</b>             | VDC | 24                 |
| <b>CI</b>                          |     | RS485              |

### 3.2 External Dimensions and Installation Dimensions

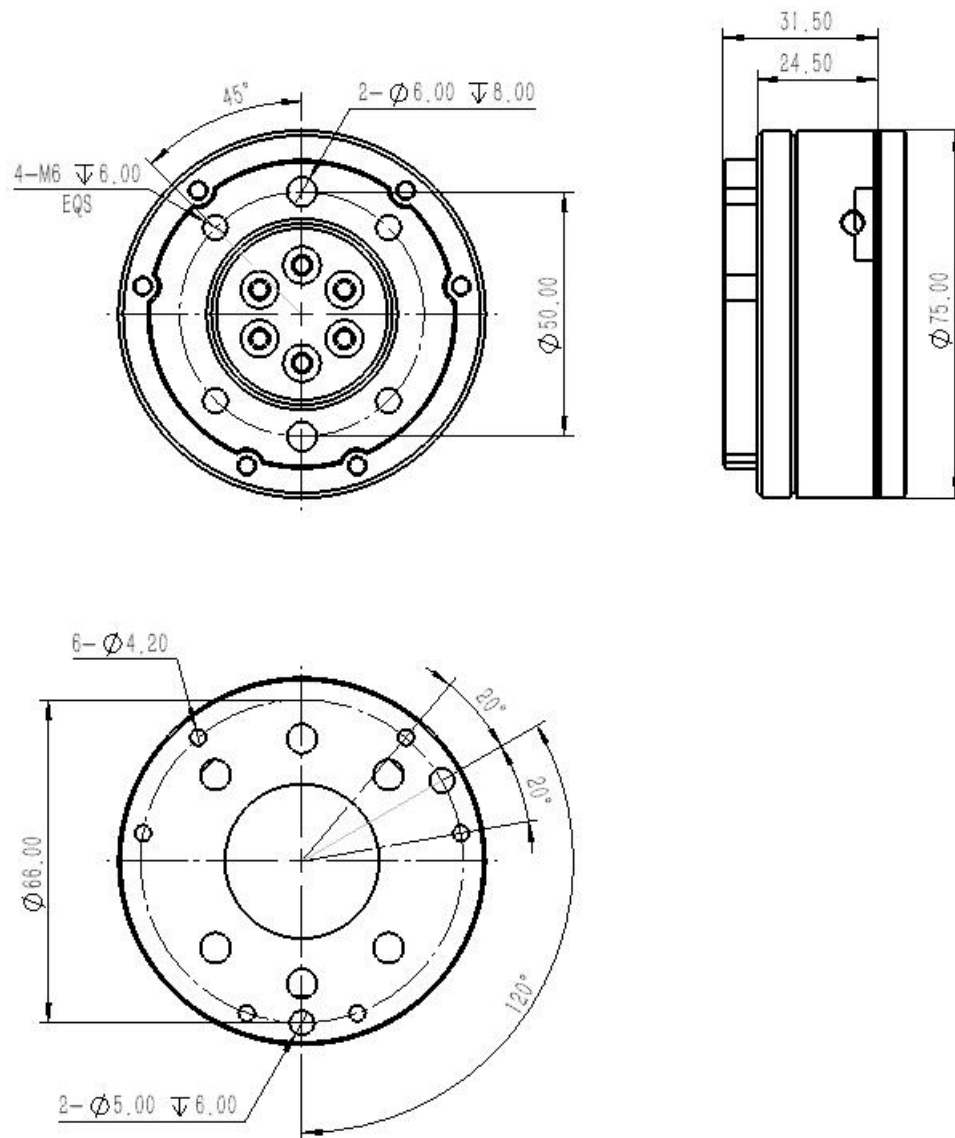


Figure 6: External Dimensions and Mounting Dimensions of FA7155

## 4. Protocol

### 4.1 Communication Specification

| Communication parameters | Price |
|--------------------------|-------|
| Communication interface  | RS485 |

|              |                    |
|--------------|--------------------|
| Baud rate    | 115200             |
| Work pattern | Send automatically |

## 4.2 Data Format

The six-axis force sensor continuously outputs data when powered on. Each frame contains 28 bytes, including a 2-byte fixed header, six float-type data points, and a 2-byte CRC check bit.

example :

The actual 485 data frames received (only two force data points are listed here, with subsequent formats following the same format):

|                  |      |                    |    |    |    |    |    |    |    |       |               |    |
|------------------|------|--------------------|----|----|----|----|----|----|----|-------|---------------|----|
| Frame 485 starts |      | 485 frames of data |    |    |    |    |    |    |    |       |               |    |
| 0x53             | 0x54 | Fx                 |    |    |    | Fy |    |    |    | ..... | CRC-16/MODBUS |    |
|                  |      | 00                 | 00 | 48 | 41 | 00 | 00 | 58 | 41 | ..... | XX            | XX |

The collected data is parsed as (little-endian format):

|      |      |       |       |       |       |
|------|------|-------|-------|-------|-------|
| Fx   | Fy   | Fz    | Mx    | My    | Mz    |
| 12.5 | 13.5 | ..... | ..... | ..... | ..... |

Actual data unit received: force: N, torque: Nm.

### 4.3 Data Read: CRC Check Example (Python) Function Command

```
def calc_crc(self, data): # 该函数用于快速查表进行crc-16/modbus计算
    crc_init = 0xFFFF
    crc_table = [
        0x0000, 0xC0C1, 0xC181, 0x0140, 0xC301, 0x03C0, 0x0280, 0xC241,
        0xC601, 0x06C0, 0x0780, 0xC741, 0x0500, 0xC5C1, 0xC481, 0x0440,
        0xCC01, 0x0CC0, 0x0D80, 0xCD41, 0x0F00, 0xCFC1, 0xCE81, 0x0E40,
        0x0A00, 0xCAC1, 0xCB81, 0x0B40, 0xC901, 0x09C0, 0x0880, 0xC841,
        0xD801, 0x18C0, 0x1980, 0xD941, 0x1B00, 0xDBC1, 0xDA81, 0x1A40,
        0x1E00, 0xDEC1, 0xDF81, 0x1F40, 0xDD01, 0x1DC0, 0x1C80, 0xDC41,
        0x1400, 0xD4C1, 0xD581, 0x1540, 0xD701, 0x17C0, 0x1680, 0xD641,
        0xD201, 0x12C0, 0x1380, 0xD341, 0x1100, 0xD1C1, 0xD081, 0x1040,
        0xF001, 0x30C0, 0x3180, 0xF141, 0x3300, 0xF3C1, 0xF281, 0x3240,
        0x3600, 0xF6C1, 0xF781, 0x3740, 0xF501, 0x35C0, 0x3480, 0xF441,
        0x3C00, 0xFCC1, 0xFD81, 0x3D40, 0xFF01, 0x3FC0, 0x3E80, 0xFE41,
        0xFA01, 0x3AC0, 0x3B80, 0xFB41, 0x3900, 0xF9C1, 0xF881, 0x3840,
        0x2800, 0xE8C1, 0xE981, 0x2940, 0xEB01, 0x2BC0, 0x2A80, 0xEA41,
        0xEE01, 0x2EC0, 0x2F80, 0xEF41, 0x2D00, 0xEDC1, 0xEC81, 0x2C40,
        0xE401, 0x24C0, 0x2580, 0xE541, 0x2700, 0xE7C1, 0xE681, 0x2640,
        0x2200, 0xE2C1, 0xE381, 0x2340, 0xE101, 0x21C0, 0x2080, 0xE041,
        0xA001, 0x60C0, 0x6180, 0xA141, 0x6300, 0xA3C1, 0xA281, 0x6240,
        0x6600, 0xA6C1, 0xA781, 0x6740, 0xA501, 0x65C0, 0x6480, 0xA441,
        0x6C00, 0xACC1, 0xAD81, 0x6D40, 0xAF01, 0x6FC0, 0x6E80, 0xAE41,
        0xAA01, 0x6AC0, 0x6B80, 0xAB41, 0x6900, 0xA9C1, 0xA881, 0x6840,
        0x7800, 0xB8C1, 0xB981, 0x7940, 0xBB01, 0x7BC0, 0x7A80, 0xBA41,
        0xBE01, 0x7EC0, 0x7F80, 0xBF41, 0x7D00, 0xBDC1, 0xBC81, 0x7C40,
        0xB401, 0x74C0, 0x7580, 0xB541, 0x7700, 0xB7C1, 0xB681, 0x7640,
        0x7200, 0xB2C1, 0xB381, 0x7340, 0xB101, 0x71C0, 0x7080, 0xB041,
        0x5000, 0x90C1, 0x9181, 0x5140, 0x9301, 0x53C0, 0x5280, 0x9241,
        0x9601, 0x56C0, 0x5780, 0x9741, 0x5500, 0x95C1, 0x9481, 0x5440,
        0x9C01, 0x5CC0, 0x5D80, 0x9D41, 0x5F00, 0x9FC1, 0x9E81, 0x5E40,
        0x5A00, 0x9AC1, 0x9B81, 0x5B40, 0x9901, 0x59C0, 0x5880, 0x9841,
        0x8801, 0x48C0, 0x4980, 0x8941, 0x4B00, 0x8BC1, 0x8A81, 0x4A40,
        0x4E00, 0x8EC1, 0x8F81, 0x4F40, 0x8D01, 0x4DC0, 0x4C80, 0x8C41,
        0x4400, 0x84C1, 0x8581, 0x4540, 0x8701, 0x47C0, 0x4680, 0x8641,
        0x8201, 0x42C0, 0x4380, 0x8341, 0x4100, 0x81C1, 0x8081, 0x4040
    ]
    crc = crc_init
    for byte in data:
        crc = (crc >> 8) ^ crc_table[(crc ^ byte) & 0xFF]
    return crc

def _handle_request(self): # 该函数用于进行数据接收解析
    while self.receive_buffer: # 当接收缓冲区不为空
        if self.receive_buffer[0] == 0x53 and self.receive_buffer[1] == 0x54: # 六维力起始帧 (2字节)
            if len(self.receive_buffer) >= 28: # 判断接收数据帧大于28个字节
                crc = self.receive_buffer[-2:] # 取帧尾两字节 (帧crc)
                integer = int.from_bytes(crc, byteorder='little') # 将字节转成int类型
                if integer == self.calc_crc(self.receive_buffer[:-2]): # 取帧前26字节进行crc计算(包含帧头), 并判断是否与帧尾两字节值相同
                    """
                    校验成功,则将帧中的24字节进行拼接
                    拼接顺序为: fx,fy,fz,mx,my,mz
                    """
                    msg = self.receive_buffer[2:26]
                    dofs = list(msg)
                    parsed_data = []
                    for j in range(6):
                        channel_bytes = data[j*4 : (j+1)*4]
                        value = struct.unpack('<f', bytes(channel_bytes))[0]
                    self.receive_buffer = self.receive_buffer[28:]
                else:
                    break
```

Note: The default baud rate for a six-axis force sensor upon power-on is 115200.

## 5. Maintenance

For mechanisms with frequent reciprocating motion, sensors may develop bolt loosening or cable bending damage during operation. Regularly inspect sensor connection bolts for tightness and cables for proper contact or damage. If bolts loosen, replace them under power-off conditions and verify sensor thread integrity. Replace damaged cables by contacting the company.

If the sensor surface is covered with dust and dirt, it is recommended to wipe the sensor surface with an alcohol swab. The sensor must be powered off during wiping. The sensor should be avoided in high-temperature, high-humidity, or dusty environments to prevent liquid and dust from entering and affecting its performance.

For high-precision or full-scale applications, the sensor must be recalibrated regularly to ensure its accuracy.

## 6. Usage Notes

- (1) To ensure the accuracy of the sensor, it should be avoided in high temperature and humidity, drastic temperature change, high impact, and dust.
- (2) Ensure the sensor is properly packaged before transportation. During transit, avoid placing heavy objects on the packaging and prevent shock loads exceeding the sensor's measurement range.
- (3) The sensor is designed to resist overload, but it can only ensure that the sensor will not be damaged under certain overload in extreme cases. The sensor should be used in the specified range when single-axis loading and composite loading are used. Long-term use beyond the range will affect the life of the sensor and its measurement accuracy.
- (4) When installing sensor bolts, first screw all bolts in without tightening. After all bolts are inserted, follow the principle of "counter-tightening" to sequentially tighten the bolts, ensuring uniform force distribution across all mechanical interfaces of the sensor. If the bolts do not engage properly, do not continue applying force to avoid thread damage; instead, replace the bolts and re-tighten them.
- (5) The sensor and actuator must be de-energized when disassembling and installing the sensor.
- (6) When wiring, always follow the cable color codes to prevent signal readout failures or sensor damage caused by incorrect connections.
- (7) When a sensor is powered on for the first time or after prolonged inactivity, it must be powered on for 30 minutes before resuming use.
- (8) In order to reduce the influence of zero drift on the measurement accuracy of the sensor, the sensor needs to be reset frequently.

- (9) When operating the sensor, users should avoid damage to the internal high-precision measurement circuit caused by external static electricity. When connecting components to the sensor, proper grounding procedures must be followed.
- (10) Users must install sensors and custom tools using the fasteners specified in this manual to prevent damage to the sensor's mechanical structure caused by incompatible fasteners.
- (11) Users should avoid external impacts (such as hammering or dropping) to the sensor during use, as this may adversely affect its performance.
- (12) Users are not permitted to disassemble or modify the internal circuits and mechanical components of the sensor without authorization. Should any modifications result in sensor malfunction, the company shall not be held liable.
- (13) If any abnormal situation not mentioned in this document occurs during use, do not operate it. Contact after-sales personnel if necessary.

## 7. Product Packaging

- |                                                     |           |
|-----------------------------------------------------|-----------|
| ● FA7155 Series Sensors (with Cable)                | 1         |
| ● Adaptation bolt                                   | a surname |
| ● User Manual of FA7155 Series (Electronic Version) | Volume 1  |
| ● Product packaging box                             | 1         |

## 8. Contact Way

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Website: [www.fibossensor.com](http://www.fibossensor.com)